

Literature Review Report

Stereolithography-based Bioprinting (SLA)

Abstract: As the world population grows and more people face serious illnesses, the demand for organ transplants has surpassed the availability of donor organs. Advanced 3D bioprinting using stereolithography (SLA) meets this demand by enabling the precise fabrication of tissue scaffolds and custom implants through high-resolution UV laser curing, using a controlled layer-by-layer curing principle.

Despite UV penetration limiting the depth of cure, SLA is able to offer great biocompatibility with low-toxicity photoinitiators and reduced material waste. This literature review discusses the conventional SLA principles, the use of near-infrared (NIR) for deeper tissue penetration, stereo-thermal-lithography (STLG) for faster polymerization, and two-photon polymerization (TPP) for sub-micron resolution.

Keywords: Stereolithography (SLA); bioprinting; laser polymerization; NIR polymerization; stereo-thermal-lithography; two-photon-polymerization

1. Introduction

Laser based photopolymerization has become into quite few of different types to build more and more accurate structures that can be applied to medicine and engineering. This report talks about the main platforms that shows how this technology has changed over time: standard stereolithography (SLA), near-infrared (NIR) photopolymerization, and two-photon polymerization (2PP).

First, we will discuss the conventional SLA, which uses controlled laser and single-photon ultraviolet (UV) light to cure photocurable resins. This part covers the fundamental principle for using UV light, the things you need to use it, and the effect that can happen, like how deep it can go and how it can be phototoxic. The article goes on to explain about new NIR-based photopolymerization methods that use less energy, are better for living things, and have more depth penetration. These technique let you print with more materials types and in cases when UV light doesn't perform as effective.

The last part talks about two-photon polymerization, which is a nonlinear optical process that lets you produce very small voxels and features that are less than a micron across. 2PP print various 3D shapes without any limitations, create complicated scaffolds, and print in delicate biological environments. One of the best things about modern microfabrication and bioprinting is that it can use femtosecond lasers and nonlinear absorption to control polymerization.

The report discusses a number of things that show how laser-assisted photopolymerization has moved from older UV-based systems to newer NIR and multiphoton systems. It talks about how they work, what they are good for, what they are bad for, and how they are becoming more important in biomedical production.

2. Conventional Stereolithography

Conventional SLA is one of the earliest laser-based vat-photopolymerization techniques used to solidify liquid photopolymers through UV exposure. It operates by “localized photopolymerization triggered by UV radiation” [1] [2] [3] [4]. Because the process cures material with high spatial accuracy, SLA has remained important in biomedical fabrication where fine structural control is required.

In bioprinting, SLA is valued for its ability to form microscale features and smooth surfaces using controlled laser scanning. Reviews on photopolymerizable biomaterials explain that polymerization relies on monomers or oligomers that solidify when exposed to a suitable light wavelength [5]. This mechanism relies on the function of photoinitiators with the laser source, create optical control is critical to print quality.

However, conventional SLA systems have biological constraints. Some studies mention that UV exposure and radical formation can influence cell viability, especially under high fluence or prolonged irradiation [4]. These challenges have pushed the use of modified exposure strategies and more biocompatible photoinitiator systems.

Overall, SLA remains most used for laser-assisted bioprinting because of the optical performance and high resolution, even as research continues to improve photoinitiator chemistry and wavelength selection for better biological compatibility.

2.1 Fundamental Principles of Conventional SLA

Conventional stereolithography forms 3D structures through a layer-by-layer solidification of photopolymer resin using a focused ultraviolet laser. The process begins with a liquid resin composed of monomers, oligomers, photoinitiators, and additives. Photopolymerization reviews describe this as the use of “liquid monomers/oligomers that can be cured upon exposure to a light source of specific wavelength to form thermosets” [3]. During printing, the UV laser scans across the resin surface to define each cross-section of the model. In the journal written by Huang et al. note that SLA “scans a focused laser beam to cure the irradiated resin” [2], forming a solid layer that adheres to the previously cured one. The platform then moves to allow fresh resin to flow over the printed area before the next exposure cycle begins.

The way of the resin hardens depends on the laser energy distribution and resin absorption. Key photopolymerization parameters include the **critical exposure (E_c)** required to start polymerization and the **penetration depth (D_p)** that determines how far the light travels before its intensity drops below this threshold. Photopolymerization study say that E_c and D_p are important to predict the cure depth will go under a Gaussian

laser beam [6]. These parameters control layer thickness, bonding strength, and the degree of feature overlap between scan lines.

Because the laser beam has an inherent Gaussian intensity profile, the cured region takes on an ellipsoidal “voxel” shape rather than a perfect cylinder, and the highest intensity at the beam center produces deeper curing while the falling Gaussian edges limit lateral resolution [7]. Exposure time, scan speed, and resin absorption further determine voxel geometry and scaling [6] [7].

Overall, the working principle of SLA is defined by the interaction of UV laser energy with a photoreactive resin, combined with precise mechanical movement of the platform. The interplay between E_c , D_p , voxel geometry, and controlled scanning allows SLA to achieve consistent layer formation and high fidelity in complex geometries.

2.2 Laser Physics & Light-Material Interaction in SLA

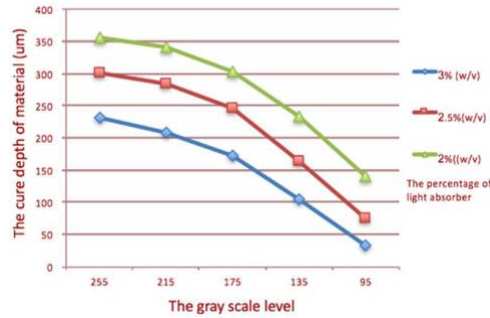


Figure 1: The impact result of curing depth, grayscale levels of input light, and different light absorbers [6]

Figure 1 Show that laser operation is central to the resolution and accuracy of conventional stereolithography, particularly in bioprinting where precise energy delivery affects cure depth, voxel geometry (gray scale area), and material compatibility [6] [7]. SLA systems commonly use near-UV lasers such as 325 nm HeCd or 355 nm DPSS sources because these wavelengths match the absorption spectra of Type I photoinitiators [5]. Reviews on photopolymerizable biomaterials explain that efficient curing requires the laser wavelength to coincide with photoinitiator absorption bands so that radicals are generated effectively [3] [4].

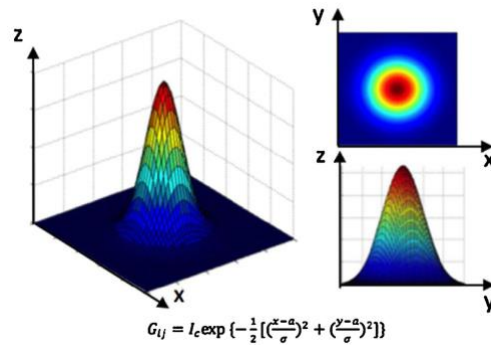


Figure 2: An illustration of Gaussian function for light energy distribution of a pixel in the projection image [6]

The laser beam used in SLA typically follows a TEM₀₀ Gaussian mode, as seen in Fig. 2 that the highest intensity appears at the center of the beam and gradually decreases toward the edges [7]. Optical studies report that UV systems used in SLA display Gaussian profiles that strongly influence the curing region and lateral resolution [6] [7]. When the beam is focused through an objective lens, the resulting spot size is determined by the wavelength, the numerical aperture of the optics, and the beam quality factor M^2 [7]. A low M^2 value shows a beam shape close to the ideal Gaussian condition, allowing smaller focal spots and improved feature sharpness [7].

Laser penetration to the resin follows the Beer-Lambert law, which effect to the exponential intensity decay with depth [3] [6]. Photopolymerization literature notes that the critical exposure E_c and the penetration depth D_p are essential parameters for predicting how deeply the resin cures under a Gaussian beam [3] [6]. Materials with high absorption result in shallower cure depths and support thin layers with high resolution [6] [7]. Materials with lower absorption allow deeper curing but reduce vertical precision [6] [7].

The interaction between the Gaussian beam and the resin creates an ellipsoidal voxel rather than a cylindrical cured region [6] [7]. High-resolution SLA studies describe that the Gaussian energy distribution forms a voxel that is deepest at the beam center and tapers outward where the energy falls below the critical exposure value [7]. The voxel shape determines how much scan-line overlap is needed for continuous layers. Low overlap causes weak bonding or voids. Excessive overlap leads to overcure and dimensional inaccuracies [7].

The thermal impact during SLA is small because of UV lasers at operate at low powers [5]. However, the localized temperature grows can still change resin viscosity and effect the polymerization rates [3]. Fast scanning lowers the thermal buildup but may underexpose deeper regions. Slow scanning increases local heating and may accelerate radical formation [3] [5]. A good calibration of laser power and scanning speed is important for stable curing, especially in bioprinting where excessive radical concentration or heat can affect cell survival [3] [4].

The laser path across the resin surface is controlled using galvanometer mirrors or fixed-beam mechanisms [2] [5]. Galvanometer systems allow rapid scanning but can introduce focal variation at larger angles, which slightly enlarges the spot size [2]. Analyses of SLA systems explain that consistent beam quality across the build plane requires calibration and optical correction [2]. Fixed-beam systems avoid this issue by moving the stage instead of the beam, although their scanning speed is usually slower [2].

So, the laser physics implied in stereolithography, including the selection of wavelength, the Gaussian beam shape, optical attenuation, and voxel formation, directly influences the resolution, uniformity, and biocompatibility of SLA prints. These optical principles form the base for optimizing SLA in both structural manufacturing and bioprinting applications.

2.3 Photopolymerization Chemistry

Photopolymerization is the chemical mechanism that enables liquid resin in stereolithography to convert into a solid structure when exposed to ultraviolet light. The reaction starts when photoinitiator molecules absorb photons and form reactive radicals [3]. Reviews on photopolymerization explain that 3D curing depends on photoactive monomers and oligomers that crosslink when irradiated at an appropriate wavelength [3] [8].

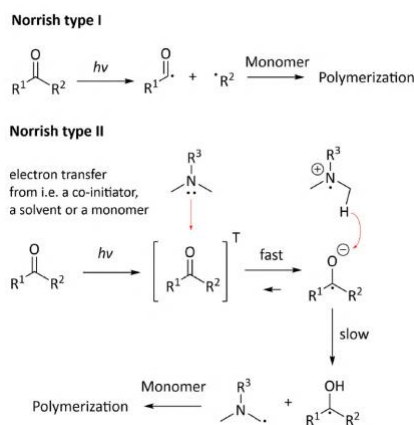


Figure 3: Type 1 photoinitiators (Norrish type 1) and Type 2 photoinitiators (Norrish Type 2)

In most conventional SLA systems, the chemistry follows a free-radical chain reaction. The first stage is initiation, where a photoinitiator absorbs UV light and transitions into an excited state [3]. Photopolymerization studies describe how Type I photoinitiators undergo rapid cleavage to produce two radical fragments that begin polymer chain growth [3] [8]. These radicals attach to acrylate or methacrylate monomers, forming the growing polymer chain. This marks the propagation stage [3]. Propagation continues very quickly as long as light intensity remains above the critical exposure needed to sustain radical formation [3].

Termination happens when two radical chains combine or when a radical becomes trapped in the network [3] [8]. The balance between initiation, propagation, and termination determines the final conversion of the resin and impact to the mechanical properties such as stiffness and shrinkage [3].

Oxygen in the resin can inhibit polymerization by reacting with free radicals [3]. This forms a thin layer near the surface where curing is slower and once radicals accumulate beyond the inhibition threshold, the reaction proceeds normally [3]. Photopolymerization sources emphasize that oxygen inhibition must be considered in low exposure or cell-laden systems because the presence of dissolved oxygen affects radical concentration [3] [8].

In bioprinting applications, the chemistry is strongly linked to cell compatibility. UV-induced radicals can interact with biological molecules, and studies note that excessive radical concentration or prolonged UV exposure may reduce cell viability [4] [8] [9]. This makes the selection of photoinitiator type and concentration important when cells are present in the bioink [4] [8].

Overall, photopolymerization chemistry dictates how the resin responds to optical energy. The interaction between photoinitiators, monomer structure, oxygen inhibition, and exposure conditions determines cure depth, crosslink density, and suitability for use in bioprinting environments.

2.4 Process Workflow and Optical Control

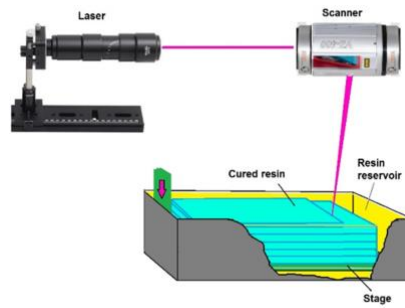


Figure 4: Schematic illustration of typical SLA processes [10]

The stereolithography adds mechanical motion, optical exposure, and photopolymerization into a continues process that builds a 3D object layer by layer [2] [5]. The model is first converted into thin slices that create the paths followed by the ultraviolet laser [5]. Stereolithography systems use a focused laser beam to scan and cure each layer of the resin [2]. After a layer is formed, the platform moves to allow fresh liquid resin to cover the previous layer before the next exposure step.

The laser scanning strategy influences accuracy and curing uniformity [2]. Most systems use galvanometer mirrors that redirect the beam rapidly across the resin surface [2]. The deflection angle of these mirrors affects beam focus and can slightly change the spot size away from the optical center [2]. Fixed-beam systems avoid this variation by moving the build stage mechanically, although their scanning speed is usually lower [2].

Each exposure forms a voxel shaped by the Gaussian intensity profile of the laser and by the absorption characteristics of the resin [6] [7]. Adjacent voxels must overlap sufficiently to form a continuous layer. Too little overlap leads to weak bonding, while excessive overlap can increase overcure [7]. The spacing between scan lines, commonly called hatch spacing, is therefore an important parameter in optical control. Hatch spacing and exposure dose determine the consistency of each layer and the dimensional stability of the part [6] [7].

Resin recoating is another factor in the workflow. After each layer is cured, the resin must flow smoothly to create a uniform liquid surface for the next exposure [2] [5]. The recoating depends on resin viscosity, platform speed, and the mechanical design of the system [5].

Post curing is typically performed after printing to increase polymer conversion and strengthen the final structure. Evenly distributed UV exposure during post-curing ensures that the monomers or initiators that not

reacted within the printed part are converted more completely, which improves mechanical properties and reduces residual stress [2] [5].

Overall, the stereolithography workflow requires coordinated control of scanning speed, exposure dose, hatch spacing, platform movement, and recoating behavior to regulate how the laser interacts with the resin and how each layer bonds to the next. This integration of optical and mechanical processes is essential for achieving stable curing and accurate geometry, particularly in bioprinting applications.

2.5 SLA in Bioprinting: UV Limitations and Compatibility

When stereolithography is applied to bioprinting, the use of ultraviolet light introduces important biological constraints that do not appear in acellular printing. UV irradiation can be harmful to living cells because high-energy photons may damage DNA and other cellular components [1] [4]. This limits the range of exposure doses that can be used when cells are incorporated directly into the printed material [1].

Radical formation during polymerization is another source of stress on cells. Excessive radical concentration can reduce cell viability, particularly when photoinitiator levels are high or when exposure times are long [4] [9]. These effects must be controlled by careful adjustment of photoinitiator concentration, laser fluence, and scanning speed to avoid producing radicals faster than cells can tolerate [9].

Optical penetration depth also becomes a limiting factor in bioprinting. Biological hydrogels frequently absorb or scatter UV light more strongly than synthetic resins. Reduced penetration limits the curing of thick cell-laden constructs and increases the risk of incomplete crosslinking in deeper regions [10] [11]. As a result, most SLA bioprinting is limited to relatively thin structures or requires modified formulations to improve optical clarity [11].

UV exposure also can cause local heating, although the thermal effects in SLA are normally small. Even slight temperature increases may influence cell metabolism or protein stability. For this reason, bioprinting normally use lower laser powers, shorter exposure durations, or modified scanning strategies to limit thermal accumulation [4] [8].

Together, these limits require that SLA bioprinting be carefully improved to balance curing performance with biocompatibility. Selection of initiators with improved biocompatibility, adjustments of exposure parameters, and the use of visible or near-infrared alternatives are common strategies to reduce UV-related stress while maintaining adequate resolution.

3. Near-Infrared (NIR) Stereolithography

In comparison with Conventional Stereolithography, near infrared stereolithography employs infrared radiation (IR) which has a few benefits where one of them is better depth penetration. The penetration depth of

light in a material is inversely related to its energy and how easily it is scattered and absorbed. Higher energy photons like UV are more readily absorbed by molecules, causing electronic excitation and breaking bonds [12]. Conversely within the NIR-spectrum, there is an “optical window” where key tissue chromophores (light-absorbing molecules) like hemoglobin, melanin, and water have relatively low absorption coefficients in this range, enabling better depth penetration [13]. NIR is also safer as high-energy blue/violet visible light has been linked to retinal damage and can cause photochemical reactions in the skin. UV light, particularly UV-B and UV-C, is a known mutagen and carcinogen. It carries enough energy to directly damage DNA and cause skin cancer. UV-A can cause premature skin aging and contributes to skin cancer risk [14].

For the polymerization of monomer with NIR Stereolithography, a photoinitiating system (PIS) is required where light energy is converted into chemical energy in the form of reactive species such as free radicals [15]. For NIR, PIS is classified into two categories which are cyanine-based system and upconversion nanoparticle system, both representing Type-II photoinitiator. Heptamethine cyanines serve as the primary NIR absorbers, exhibiting maximum absorption wavelengths around 783-823 nm shown in Figure 5 with exceptionally high molar extinction coefficients (ϵ) exceeding $3 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$ shown in Figure 6 [16]. This high ϵ is the property that enables them to efficiently absorb NIR light for deep penetration in SLA. To leverage this property effectively, the dye molecules must be homogeneously dispersed at a molecular level to ensure uniform absorption and initiation throughout the resin volume, preventing scattering and shadowing. Achieving this homogeneous dispersion requires dissolving the solid dye powder into a suitable medium, which is normally used with a compatible solvent, e.g. anhydrous dichloromethane, ethyl acetate that is either evaporated after dispersion or is present as a minor component of the final resin formulation.

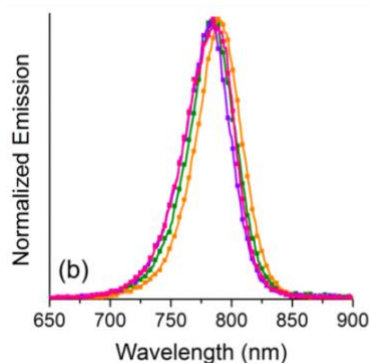


Figure 5: Maximum absorption wavelength of heptamethine cyanine

Cyanine	Solvent	λ_{abs}	ϵ
6a	Ethyl acetate	778	197,000
	Dichloromethane	789	288,000
	Methanol	774	205,000
	Acetonitrile	775	225,000
6b	Ethyl acetate	783	189,000
	Dichloromethane	789	206,000
	Methanol	780	365,000
	Acetonitrile	780	284,000

Figure 6: Heptamethine cyanine with high coefficient of extinction in different solvents

These cyanines function through photoinduced electron transfer (PET) mechanisms when combined with iodonium salts as co-initiators. The polymerization reaction relies on a donor-acceptor interaction and is separated into four different phases represented by excitation, transfer, radical generation and finally bleaching. Upon NIR excitation, the cyanine molecule transitions to an excited state. An electron is transferred to an

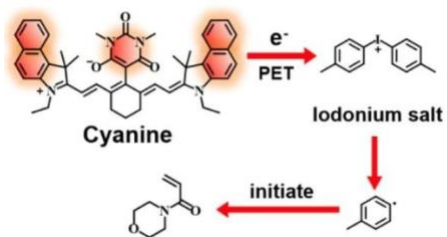


Figure 7: PET Processes

Iodonium salt which acts as the co-initiator. This produces an aryl radical that initiates polymerization. This process destroys the cyanine dye, which is known as the bleaching effect and prevents the cyanine from interfering with the polymerization process [17]. Figure 7 shows the principle of photoinduced electron transfer with figure x+1 detailing the exact elements during the reaction process.

For producing 808 nm light diode lasers are overwhelmingly the preferred choice for industrial applications for multiple reasons. Diode lasers serve as an optimal excitation source for NIR photopolymerization due to their emission of monochromatic light, specifically at a peak wavelength of 808 nm. This wavelength aligns precisely with the absorption spectrum of heptamethine cyanine sensitizers—typically situated between 783 and 823 nm—ensuring high selectivity in initiating the photochemical reaction [18]. By operating in continuous wave (CW) mode, the laser facilitates sustained energy delivery, which is critical for maintaining controlled tissue heating without thermal damage [19]. Furthermore, when tunable within a high-intensity range of 0.2 to 2.4 W cm⁻², the system effectively triggers the photothermal heating mechanism required to drive polymerization [20].

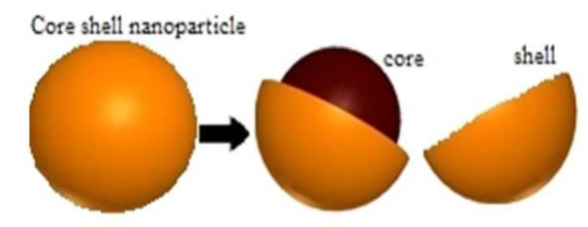


Figure 8: Core-shell architecture for UCNPs

For Upconversion Nanoparticle-Mediated System, Upconversion nanoparticles (UCNPs) convert NIR light into UV/visible emissions through a multi-photon stepwise process. This system is introduced with a core-shell architecture as shown in Figure 8 with NaYF₄ doped with Yb³⁺ and Er³⁺ or Tm³⁺

acting as the core and the shell is undoped NaYF₄ nanoparticles. In upconversion, the nanoparticle must sequentially absorb two or more low-energy NIR photons (usually 980 nm) to emit a single high-energy UV/visible photon. UCNPs (core-only) are too dim for 3D printing because they lose up to 90%+ of their energy to surface quenching. Therefore, with this architecture, the shell acts as a physical barrier, sealing the active ions inside and preventing energy loss [21].

The main components involved in UCNP-assisted

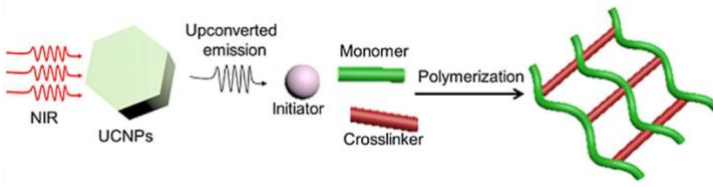


Figure 10: Photopolymerisation of UCNPs

photopolymerization were monomers/crosslinkers, UCNPs, and photosensitizer/co-initiators. UCNPs convert NIR light to UV or visible light. The upconverted UV/Visible photons excites photosensitizers, and the excited photosensitizers/photoinitiators can subsequently initiate polymerization of monomers and cross-linkers as shown in Figure 9 simplified in Figure 10.

Specifically, Tm³⁺ + /Er³⁺ -containing UCNPs are two kinds of classical upconverting phosphors and they are mostly used in UCNP-assisted photopolymerization. UV emissions of Tm³⁺ -containing UCNPs are employed for UV photo-sensitizers. Visible photosensitizers can absorb the blue emissions of Tm³⁺ -containing UCNPs (Figure 11 left) or the green or red emissions of Er³⁺ -containing UCNPs (Figure 11 right) in Figure 11.

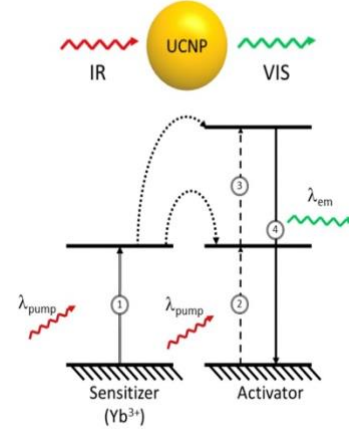


Figure 9: Upconversion Process

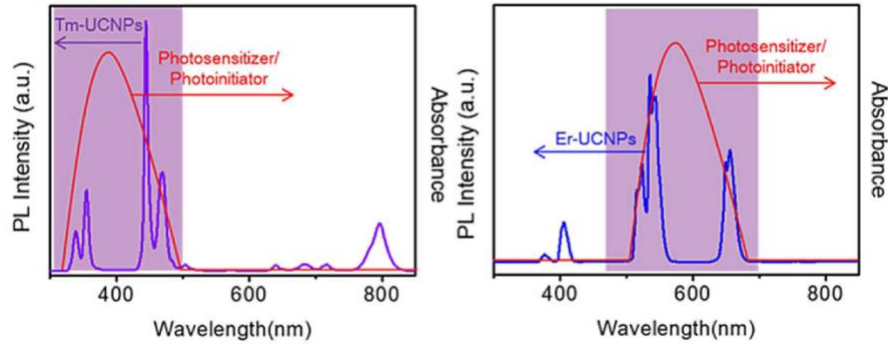


Figure 11: Visible photosensitizers can absorb the blue emissions of Tm³⁺ -containing UCNPs (Figure 11 left) or the green or red emissions of Er³⁺ -containing UCNPs (Figure 11 right)

Upconversion is a non-linear process. The emission intensity ($I_{emission}$) is proportional to the excitation power (P) raised to the power of the number of photons (n):

$$I_{emission} \propto P^n$$

Polymerization only happens at the very focal point of the laser where intensity is highest. The surrounding "halo" of the laser beam is too weak to trigger upconversion. This drastically sharpens the voxel (volumetric pixel), allowing for sub-diffraction features (<200 nm resolution) like expensive Two-Photon Polymerization (TPP) systems, but using cheap Continuous Wave (CW) lasers. The voxel sizes are adjustable, with transverse dimensions ranging from 1.3 to 2.8 μm and axial sizes from 7.7 to 59 μm by adjusting NIR power from 7.7×10^4 to 1.1×10^6 W cm⁻². This feature enables decoupling of feature size from polymerization degree, offering unique optimization opportunities [22].

Polymerization only happens at the very focal point of the laser where intensity is highest. The surrounding "halo" of the laser beam is too weak to trigger upconversion. This drastically sharpens the voxel (volumetric pixel), allowing for sub-diffraction features (<200 nm resolution) like expensive Two-Photon Polymerization (TPP) systems, but using cheap Continuous Wave (CW) lasers. The voxel sizes are adjustable, with transverse dimensions ranging from 1.3 to 2.8 μm and axial sizes from 7.7 to 59 μm by adjusting NIR power from 7.7×10^4 to $1.1 \times 10^6 \text{ W cm}^{-2}$. This feature enables decoupling of feature size from polymerization degree, offering unique optimization opportunities [22]. The 980 nm continuous-wave (CW) diode laser is widely recognized as the optimal excitation source for upconversion nanoparticle (UCNP)-assisted near-infrared (NIR) stereolithography due to its strong spectral alignment with the absorption peak of Yb^{3+} sensitizer ions. Multiple studies confirm that UCNPs such as $\text{NaYF}_4:\text{Yb,Er}$ or Yb,Tm absorb most efficiently at this wavelength, enabling effective upconversion [23]. Furthermore, these systems operate efficiently at relatively low power densities, typically in the 10–100 W/cm^2 range, sufficient to induce UV or visible upconversion needed for polymer curing. The 980 nm wavelength also exhibits exceptional penetration in photocurable resins, with experimental evidence showing photopolymerization depths of up to 10–13.7 cm—far surpassing conventional UV-based systems. This deep curing capability allows for volumetric 3D printing, enabling direct writing at any point within the resin volume, unconstrained by traditional layer-by-layer techniques. Finally, commercial 980 nm CW diode lasers are built for stable, extended operation, often running continuously for 5 or more hours without degradation, making them ideal for long-duration printing applications [24].

4. Stereo-Thermal-Lithography (STLG)

Despite conventional SLA remaining dominant in rapid bioprinting and prototyping, it still exhibits limitations in cure durability and uniformity [25]. Light interacts strongly with biological tissues, with its wavelength being a significant factor in this interaction with living organisms [26]. The intense interaction between visible light (UV) and tissues will reduce the tissue penetration depth to less than 1 mm and increase phototoxicity [26]. Alternatively, near-infrared (NIR) light, whose wavelengths range between 700 and 1700 nm, has lower energy and longer wavelengths. It also experiences far less tissue absorption and scattering in biological tissues [26]. To solve these limitations, Stereo-thermal-lithography (STLA or STLG) was first introduced by the research team at the Centre for Rapid and Sustainable Product Development at the Polytechnic Institute of Leiria, Portugal [26] [27]. In STLG, UV and near-infrared (NIR) radiation are applied simultaneously to initiate the curing process in a liquid solution containing both UV and IR photoinitiators [27]. By using a dual-beam activation technique, STLG combines the effects of UV and IR radiation to produce heat [27].

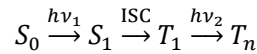
In contrast, conventional SLA relies on single-photon UV photopolymerization to activate photoinitiators and cure one layer at a time, which increases both material cost and processing time [25]. Moreover, UV exposure can damage cellular components and DNA and limit the range of compatible materials [28]. In the latest work of Jinyu Sun et al. [29], photothermal applications of STLG were proposed. Photothermal nanoparticles

transform NIR-II light (typical wavelength range: 900–1,880 nm) [30] into localized heat. This heat then activates thermal initiators to crosslink resins or bioinks in a layer-by-layer manner [31]. By replacing photoinitiators with thermal initiators and using NIR light to provide localized heating, the STL method can achieve faster polymerization, increased process efficiency, greater control over printing conditions, and lower cellular toxicity [25].

4.1 Fundamental Principles of STL

In the STL process, both UV radiation and thermal energy (produced by infrared radiation) are used with dual photo- and thermal initiators at deliberately low concentrations. Bartolo and Mitchell [25] stated that the concentrations of both initiators are chosen so that the reaction begins only when UV radiation and thermal energy are combined [25]. Individual paths, either UV or thermal effects, produce insufficient radicals for polymerization [25]. However, at temporal intersections of both effects, the radical concentration becomes sufficiently high for the polymerization [25]. Temperature simultaneously activates thermal initiators and accelerates the kinetics of photo-initiated reactions [25]. In this way, the extent of cure is increased, and no post-cure is required [25].

Swainson presented a biphotonic excitation concept [32], in which two different electromagnetic (EM) beams at different wavelengths were used in the SLA system to write within a liquid-resin vat. Swainson mentioned that the source or beam production could be any EM or other radiation source, as long as the beams are controllable and discrete [32]. The simultaneous process then triggers the local reactions. The biphotonic excitation method relies on sequential absorption of two photons by a single initiator molecule to initiate polymerization and thus requires high photon densities and precise conditions.



S is the ground (singlet) state, T is the triplet state, $h = 6.62 \times 10^{-34} \text{ m}^2\text{kg/s}$ is Planck's constant, ν is the frequency of the absorbed photon, and ISC is the abbreviation of intersystem crossing (where the molecule transitions from an excited singlet state to an excited triplet state).

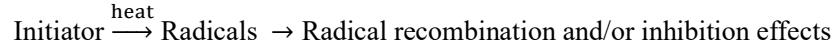
As a result, the process is slow and less efficient. The STL approach, which uses two independent polymerizations initiated when these two effects intersect, differs from Swainson's biphotonic excitation concept, offering greater efficiency through an independent initiator rather than sequential bi-photonic absorption [32].

Figure 12 shows the setup for STL presented by Bartolo's research team [27]. The system consists of a power supply, a light splitter, collimating probes, UV filters, two digital micromirror devices (DMDs), each employed with UV and IR beams, a dichroic mirror, and a polymetric system. The DMD-UV and DMD-IR can independently pattern the UV and IR beams, allowing greater control over the light's wavelength and where it should be delivered. A dichroic mirror combines the UV- and IR-patterned beams by reflecting one wavelength

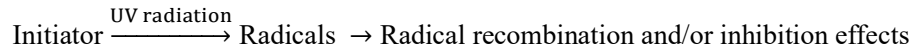
at a time while allowing another to pass through, so that both beams stay precisely aligned. In the final step, both the UV- and IR-patterned (overlapped) beams are projected onto a photopolymerizable resin. STLG triggers polymerization only where both energies intersect, thus achieving high localization and efficiency.

Bartolo and Mitchell [25] [27] presented three initiation processes in STLG:

1. Path 1 – Thermal energy effects:



2. Path 2 – UV radiation effects:



3. Path 3 – Intersection of paths 1 and 2 – Thermal energy + UV radiation effects

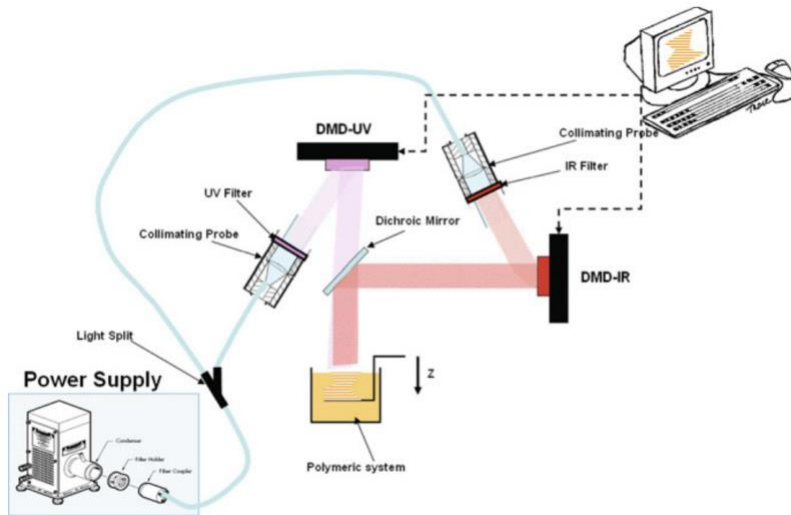
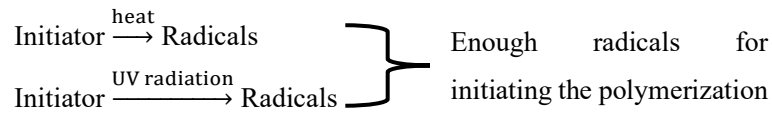


Figure 12: Irradiation process in STLG method by Bartolo [27]

Bartolo and Mitchell [25] [27] also introduced four subsystems for STLG in accordance with the initiation process. Subsystem A uses UV radiation to solidify a liquid resin that contains a certain amount of photoinitiators. Subsystem B uses the heat from IR to solidify a liquid resin that has a particular number of thermal initiators. Subsystem C corresponds to path three, which uses both heat and UV radiation to solidify a liquid resin that includes a certain amount of photoinitiators. Lastly, subsystem D corresponds to path three and is used to solidify a liquid resin containing a certain amount of both thermal and photoinitiators.

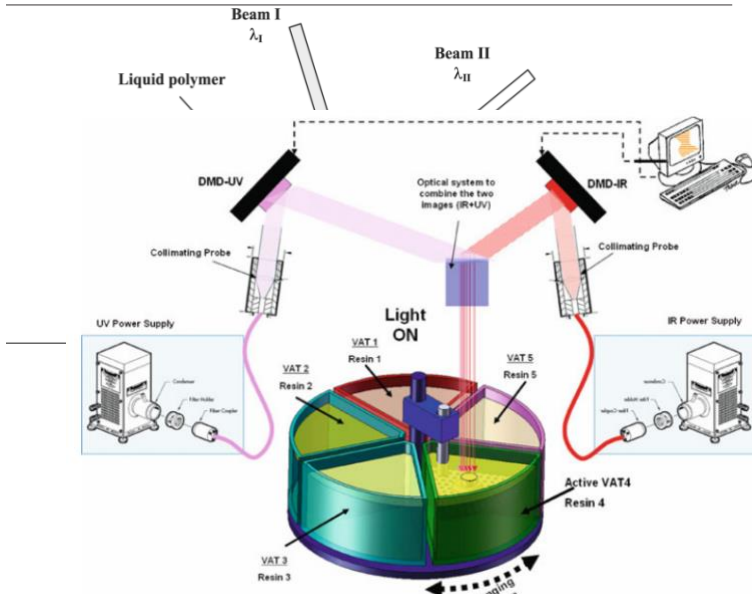


Figure 14: Multi-vat system with rotary platform

entire work cell can rotate and allows automated switching between different resins, thus enabling multi-material printing at once.

Bartolo [27] also mentioned that the multi-vat system in Figure 14 was used to produce further multi-material microscopic engineering prototypes, including nanostructures for waveguiding and photonic crystals, multi-material functional graded scaffolds for tissue engineering, other biomedical components, and micro functional metallic or ceramic parts.

Figure 15 shows the assembled multi-vat system that operates in the mask-based method [33]. The system uses subsystems A to C. A 350 W mercury lamp is used as the light source, with filters that split the UV and NIR

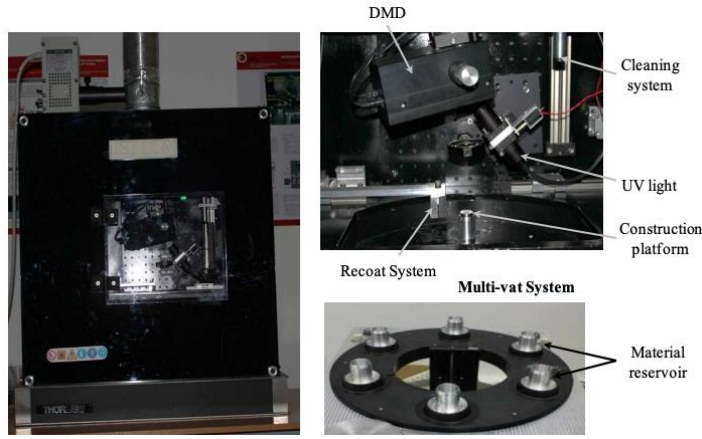


Figure 15: An overview of the STLG system assembled with its main mechanical components [22]

radiation into two wavelengths. The system uses multiple vats and incorporates vertical positioning via a MYCOSIS VT-80 stage, enabling 1 μm incremental, adjustable speeds from 0.001 to 20 mm/s.

The experimental setup included thermocouples connected to a temperature controller, heating elements, and a refrigeration system that uses cold-water flow [25]. A mask-based irradiation method was used for the initial proof-of-concept by Bartolo and Mitchell [25]. The mask contains transparent and opaque regions that can define a specific pattern to be cured into the resin [25]. In addition, the mask was applied to control the shape of the exposed region, thus allowing simultaneous irradiation with UV and thermal energy [25]. UV light that passes through the transparent parts of the mask can initiate the photopolymerization process, corresponding to the resin region [25]. Thus, mask-based irradiation is more straightforward to implement experimentally and requires less time to pattern all layers than individual point scanning used in SLA [25].

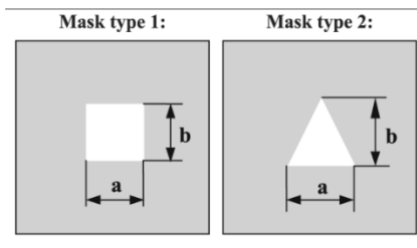


Table I Dimensions of each mask type

	Mask type 1	Mask type 2
Dimensions (mm)	$a = 11$ $b = 11$	$a = 11.5$ $b = 11$

Figure 16: Mask types used experimentally by Bartolo and Mitchell to implement STLG [25]

4.2 Technical Implementation

In the initial experiment, Bartolo and Mitchell [25] set up an experimental apparatus using an aluminum block housing with a mercury lamp (Black-Ray B100 AP, UVP Inc.), which transmitted light in the

range of 300-400 nm with $\lambda_{max} = 365.5$ a bandwidth of nm. The experimental setup included thermocouples connected to a temperature controller, heating elements, and a refrigeration system that uses cold-water flow [25]. A mask-based irradiation method was used for the initial proof-of-concept by Bartolo and Mitchell [25]. The mask contains transparent and opaque regions that can define a specific pattern to be cured into the resin [25]. In addition, the mask was applied to control the shape of the exposed region, thus allowing simultaneous irradiation with UV and thermal energy [25]. UV light that passes through the transparent parts of the mask can initiate the photopolymerization process, corresponding to the resin region [25]. Thus, mask-based irradiation is more straightforward to implement experimentally and requires less time to pattern all layers than individual point scanning used in SLA [25].

The results reported by Bartolo and Mitchell [25] show that structures fabricated using the STLG process significantly improved lateral accuracy compared to conventional SLA, with UV-only curing. According to Bartolo and Mitchell [25], the lateral dimension error measured via the STLG method was ± 0.23 mm, whereas a ± 0.56 mm error was found in the conventional process. Additionally, deeper light penetration was achieved as the initiator concentration was reduced, resulting in uniform curing and eliminating the need for post-cure treatment [25].

4.3 Advantages of STLG compared to conventional SLA

1. **Faster Polymerization:** With the combination of UV and thermal activation, STLG able to achieve a more focused curing process. The two-combined mechanisms accelerate radical formation and speed up the overall curing process.
2. **Cost-Saving:** Unlike conventional SLA, thermal initiators used in STLG are typically less expensive than photoinitiators. As a result, this reduces production costs and able to better support large-scale manufacturing.
3. **Improved Process Efficiency:** Using both UV and NIR initiators enables a more complete polymer conversion in one step, thereby minimizing the need for post-curing. Furthermore, NIR radiation penetrates deeper into the resin, improving the uniform degree of curing throughout the material.
4. **Broader Material Compatibility:** STLG supports resin that reacts to both thermal and photo-initiated reactions, provided their chemistry is compatible with the respective activation mechanisms. Nonetheless, photopolymerization is limited to materials with matching photochemistry.
5. **Reduced Cellular Toxicity:** The amount of unreacted monomers and toxic byproducts are reduced as the initiator concentrations is low and combined with the lack of extended UV post-curing. Additionally, NIR light causes minimal photodamage relative to UV light, hence improving the cell viability and function.
6. **Better Tunability:** The STLG system can be divided into four independent controllable subsystems, allowing process optimization based on the resin chemistry or application needs.
7. **Greater control over printing conditions:** The STLG approach can independently target temperature and light intensity settings, providing more precise control over cure kinetics and structural quality in printed parts.

4.4 Application in Tissue Engineering and Regenerative Medicine

High-resolution photothermal stereolithography bioprinting remains challenging because confining heat in an aqueous environment is difficult, as heat dissipates too quickly. To address the challenge, in the latest work of J. Sun *et al.* [29], a photothermal conversion mechanism was developed by applying the core principle of Imaging-Guided Microscale Photothermal Stereolithography Bioprinting (ImPSB).

The researchers aimed to achieve high-resolution aqueous bioprinting at a low cost by using thermal initiators (TIs) instead of conventional photoinitiators (PIs). The difference between ImPSB and STLG is that ImPSB aimed to achieve microscale resolution and cellular biocompatibility in aqueous photothermal SLA bioprinting, whereas the STLG method aimed to improve the accuracy, speed, and quality of macroscopic engineering prototypes by achieving higher-quality cure and avoiding post-curing.

The working principle of ImPSB is as follows:

Photothermal Initiator $\xrightarrow{\text{NIR-II Irradiation}}$ Localized Heat $\rightarrow$ Radicals $\rightarrow$ Polymerization

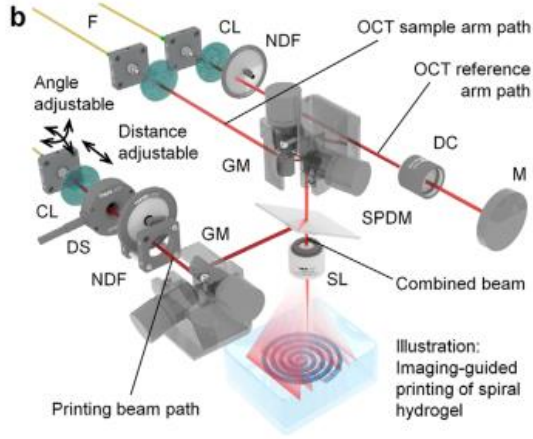


Figure 17: Schematic diagram by Jingyu Sun et al. showing the combination of two beams with the optical components [29]

J. Sun *et al.* [29] engineered a layered gold-platinum (LAuPt) nanoframework as a photothermal agent in the bioink. When exposed to NIR-II light (1100–1500 nm) for printing, these nanoframeworks absorb the NIR energy and convert it into localized heat. Firstly, the LAuPt framework was loaded with a thermal initiator. The 2,2'-azobis [2- (2-imidazolin-2-yl) propane]-dihydrochloride (AIPH) aqueous solutions were used as the thermal initiator. When exposed to NIR-II irradiation, the LAuPt absorbs the light, and the temperature rises rapidly. The surrounding medium was locally heated.

Next, the generated heat decomposes the AIPH inside the nanoframework, thus producing free alkyl radicals. These radicals then initiate polymerization of polyethylene glycol diacrylate (PEGDA) in the bioink, transforming it from a liquid to a solid hydrogel at targeted locations. The photothermal effect is confined to the illuminated focal volume by the NIR laser. As a result, the gelation is highly localized. J. Sun *et al.* [29] achieved microscale and biocompatible realization using the proposed light- and heat-polymerization STLG method. This method enables polymerization in the absence of UV light or high concentrations of photoinitiators, thereby improving cellular compatibility for bioprinting. Therefore, ImPSB is an enhanced version of STLG that uses NIR-II light instead of IR radiation.

5. Two-photon polymerization process (TPP)

Apart from the single photon polymerization, two-photon polymerization is also a key component in stereolithography, as it enables fabrication of micro- and nano- structures. As a lithography technique, TPP operates on the two-photon absorption (TPA) principle to produce fine structures ranging from hundred nanometers [34] [35] to sub-100 nm resolution [36]. TPP development began with two-photon absorption theory proposed by Göppert-Mayer in 1931, which was experimentally confirmed by Kaiser and Garrett in 1961 [37]. The TPP development progress has also been strongly driven by the laser technology advancement, since laser is essential for TPP printing. The TPP high resolution has piqued significant interest from researchers and industry to fabricate intricate microscale structures for medical application, such as microneedles [38], drug delivery system [39], and cell scaffolding [40].

5.1 TPA induced TPP

Two-photon polymerization stereolithography takes advantage of the nonlinear nature of two-photon absorption (TPA). 2PA is a third-order nonlinear process whose probability " p " scales quadratically with the light intensity " I " ($p \propto I^2$), and therefore a high intensity light source such as a laser is needed to trigger it [41]. The TPA process can be seen in Fig. 18. One-photon absorption process is a linear process where a molecule can be excited from the ground state (S_0) to an excited electronic state (S_1) if the photon energy matches the energy

gap between the two states. However, this transition can also occur when two photons having half the energy of the corresponding one-photon transition are simultaneously absorbed. In TPA, when the first photon interacts with the molecule, an intermediate and extremely short-lived virtual state ($\sim 10^{-15}$ - 10^{-16} s) is formed above the ground state as shown in Figure 18. The electronic transition will occur the moment the second photon interacts with the molecule within this short time frame. This TPA characteristic ensures the polymerization only occurs in the selected space, increasing the printing resolution.

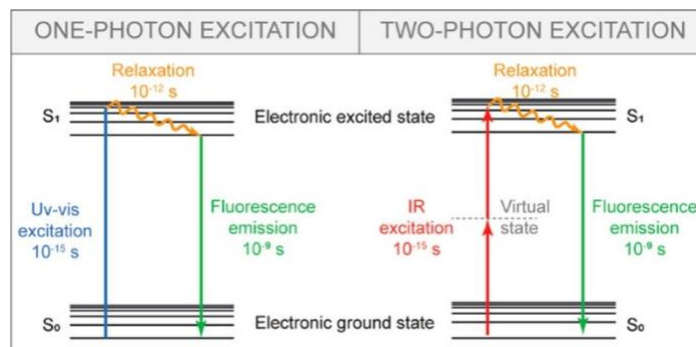


Figure 18: One-photon excitation vs two-photon excitation [41]

5.2 Laser types used in TPP and its considerations

Choosing the appropriate laser type for TPP printing is one of the most important considerations. Since TPP stereolithography requires a high-intensity light source, femtosecond lasers have become the most commonly used option. Femtosecond lasers rely on mode-locking to concentrate energy into ultrashort femtosecond pulses [42], generating the high photon density required for TPA while maintaining a low average power that reduces thermal diffusion and limits damage towards cells and biomaterials.

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In bioprinting applications, Ti:sapphire fs-pulsed lasers are the standard in most instrument setups. Meanwhile, other laser sources, such as amplified ultrafast yttrium-aluminum-garnet (YAG) lasers, second-harmonic fiber lasers, and Yb:KGW lasers, are used only in a small minority of 3D bioprinting systems. Ti:sapphire is preferred because its titanium sapphire doped lasing medium enables the construction of highly tunable lasers operating between 660 and 1100 nm, a wavelength range well suited for 3D bioprinting [43].

In TPP bioprinting, the femtosecond lasers operate mostly in the NIR range (700-800 nm). Firstly, this decision is due to the characteristic of TPA where it requires two photons with half the energy of the corresponding one-photon transition. Half the energy means that the light used will have twice the wavelength of the light used in one-photon transition. As a result, two-photon absorption occurs at red-shifted wavelengths,

most commonly in the near-infrared (NIR) and infrared (IR) region. The second reason is the narrow “transparency window” of biological tissues, which lies around 800-950 nm [44]. When higher wavelengths are used, water begins to act as a light filter. Moreover, many commonly used materials are transparent to NIR light, confining the interaction between the material and the incident light to only at the focal volume, as shown in Figure 19 [45]. In addition, using NIR light sources reduces the cytotoxicity by photodamage to cells compared with UV-visible light sources.

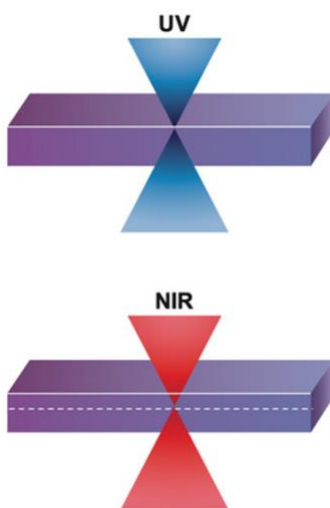


Figure 19: UV photopolymerization displaying low penetration and high surface interaction vs NIR photopolymerization displaying minimal surface interaction and increased penetration [45]

5.3 Set up and components in doing TPP stereolithography

The instrument setup for TPP stereolithography generally comprises of a laser source, galvo mirrors, focusing objective, sample holder (usually an ultrathin glass substrate), substrate, and piezoelectric stage displacements. In the past, moving the laser from the previous polymerization site to another site depended on the piezoelectric stage displacement motor, which made the process extremely slow. Adding the galvo mirrors scanning system to the setup as shown in Figure 20 enables much faster transition, expanding the microfabrication possibilities in terms of size and shape [34].

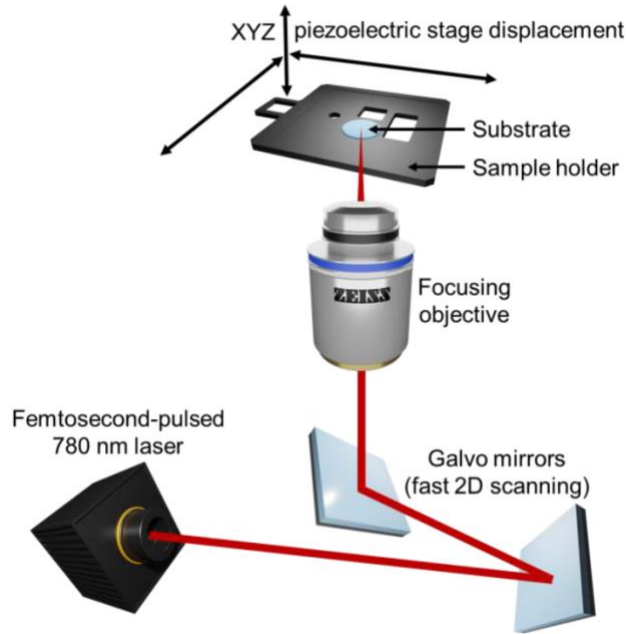


Figure 20: Two-Photon Polymerization printing setup [34]

In TPP, there are two printing configurations which are “immersion” and “DiLL” as shown in Figure 21 [34]. In immersion configuration, as shown in Figure 21a, the bioresin is placed on top of an ultrathin glass substrate while a drop of immersion oil with a refractive index matching that of the substrate is first applied between the substrate and the microscope objective. The printing starts in the interface between the substrate and the bioresin and proceeds upwards. Meanwhile, Figure 21b shows the DiLL configuration where the microscope objective is directly immersed into the bioresin which is placed at the bottom of the substrate. In this case, the bioresin also acts as the immersion medium. In DiLL configuration, the printing begins on the interface between the bioresin and the substrate and proceed downwards.

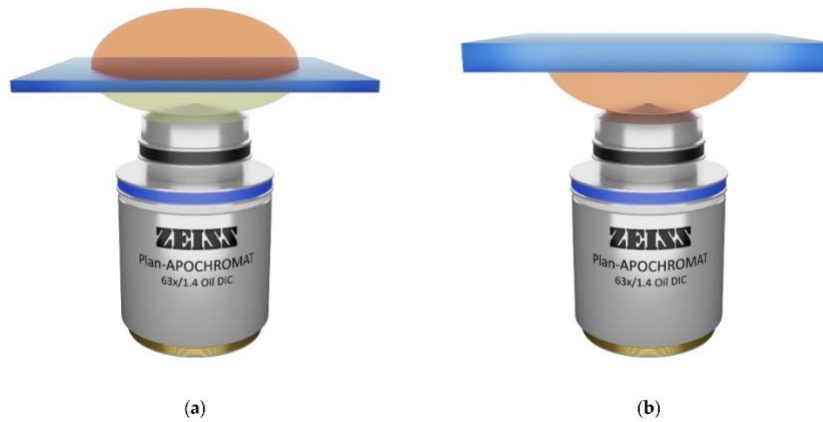


Figure 21: Oil Immersion (a) and DiLL (b) Configurations [34]

Both immersion and DiLL configurations have their own advantages and disadvantages, according to the target structures. Oil immersion configuration is best to fabricate a relatively thin and flat structures with much more details while DiLL mode is best to obtain tall structures. This comparison can be seen in Figure 22. As shown in Figure 22b, the mandala structure fabricated with DiLL configuration lacks the smallest details when compared to the oil immersion configuration. On the other hand, Figure 22d shows the pyramid wall printed with oil immersion collapsed and the pyramid tip faced deformation.

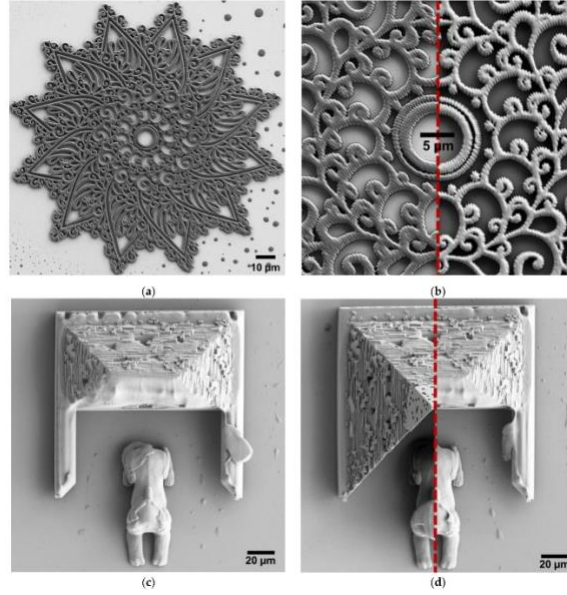


Figure 22: (a, b) Scanning electron micrographs of mandala structure (a) Printed in DiLL mode (b) Comparing the mandala structure fabricated with DiLL configuration (left) and oil immersion (right) (c, d) Scanning electron micrographs of the pyramid guarding sphynx structure (c) Printed in DiLL mode (d) Comparing the pyramid structure with DiLL configuration (left) and oil immersion (right) [34]

5.4 Choosing photoinitiator for TPP

Photoinitiator is another key in stereolithography 3D bioprinting, and the requirements for TPP photoinitiator (PI) is different with the PI for single-photon polymerization. The first requirement is a large TPA cross-section which quantifies the probability of TPA at a wavelength of interest. The PI also need to efficiently undergo intersystem crossing (ISC) to the triplet state from which radical active species can be generated to initiate polymerization [46].

The PI should also be water-soluble to dissolve well in the system. The hydrogels used in TPP bioprinting are mostly water-based hydrogels, and a water-insoluble PI will form phase separation which leads to poor resolution. Moreover, water-solubility helps in reducing cellular uptakes of PI which is toxic to the cells. Some examples of water-soluble TPP photoinitiators are G2CK, E2CK, and P2CK, which made water-soluble through the addition of carboxyl groups [47].

6. Conclusion

Stereolithography (SLA) for 3D bioprinting has gained its popularity and experienced many advancements in these recent years. The main fundamental of stereolithography revolves around the production of free radicals to initiate the polymerization. Everything starts with conventional SLA where the process depends on single photon absorption. NIR-SLA followed right after by changing the excitation light used. Instead of using UV-vis light like SLA, NIR-SLA use near infrared light that has higher tissue penetration and cause much less photodamage compared to UV light. Another SLA 3D bioprinting breakthrough is STLG which combines both UV radiation and thermal energy produced by infrared radiation to induce radical generation. STLG only requires photoinitiator and thermal initiator at low concentrations which caused less damage to the cells and biomaterials. The final advancement discussed in this paper is TPP polymerization. TPP 3D bioprinting utilizes TPA principle to induce polymerization, resulting in a very high resolution compared to the other SLA bioprinting down to sub-100 nm. Even though abundant progress has been made in SLA 3D bioprinting, further studies are needed to tackle the challenges experienced by each SLA 3D bioprinting techniques.

7. Author's Contribution

All authors contributed to the conception and refinement of the ideas presented in this literature review report. Florence Wong drafted the abstract, stereo-thermal-lithography (STLG) section and finalized the formatting; Guo Bin Djaip drafted the introduction and conventional stereolithography section; Lim Jun Heng drafted the near-infrared (NIR) stereolithography section; and Rangga Fokta proposed the topic and drafted the two-photon-polymerization section and conclusion.

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